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A Quantitative Comparison of the Parts of the Brains of Two Australian Marsupials and some Eutherian Mammals

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Abstract. The size and composition of the brains of the quokka wallaby *(Setonix brachyurus)* and the brush-tailed possum *(Trichosurus vulpecula)* were compared with those of the rat, rabbit, cat, sheep, and human. Separate structures were taken from the brains by standardized dissection and their weights compared. Particular attention was paid to the relationships between the size of the amygdala and other brain structures.

Marsupials were within the eutherian range in the ratio of whole brain weight to body weight. They were also within eutherian ranges in the proportion of the total brain weight formed by all structures, except the striatum, and in the proportion of the non-neocortical brain weight contributed by all structures but the olfactory bulbs. In all measures the marsupials were within the general range for eutherian mammals reported in the literature. The size of the amygdala was related significantly to that of the septum, hypothalamus, thalamus, lower brain stem, and olfactory bulbs.

Through approximately 70 million years of separate evolution, Australian marsupials have adapted efficiently to a large number of ecological niches similar to those occupied by eutherian mammals in other parts of the world. For this reason, the brains of Australian marsupials have been of particular interest to those neuroanatomists who attempt to relate brain structure to function.

Since *Elliot Smith* [1894] marsupial brains have been considered small relative to those from equivalently sized eutherians. They have also been considered to differ in composition from eutherian brains, having a greater proportion of rhinencephalon, but less 'neopallium' [see, e.g., *Jerison*, 1973; *Young*, 1962]. A few authors, including *Tyndale-Biscoe* [1973] have questioned the suggested morphological differences between the brains of eutherians and those of marsupials, but no quantitative studies appear to have been made which could settle the issue.

This study was designed particularly to determine whether the brain of the quokka, a small southwest Australian coastal wallaby, differs in size and composition from equivalent eutherian brains. Emphasis has been placed on the relative

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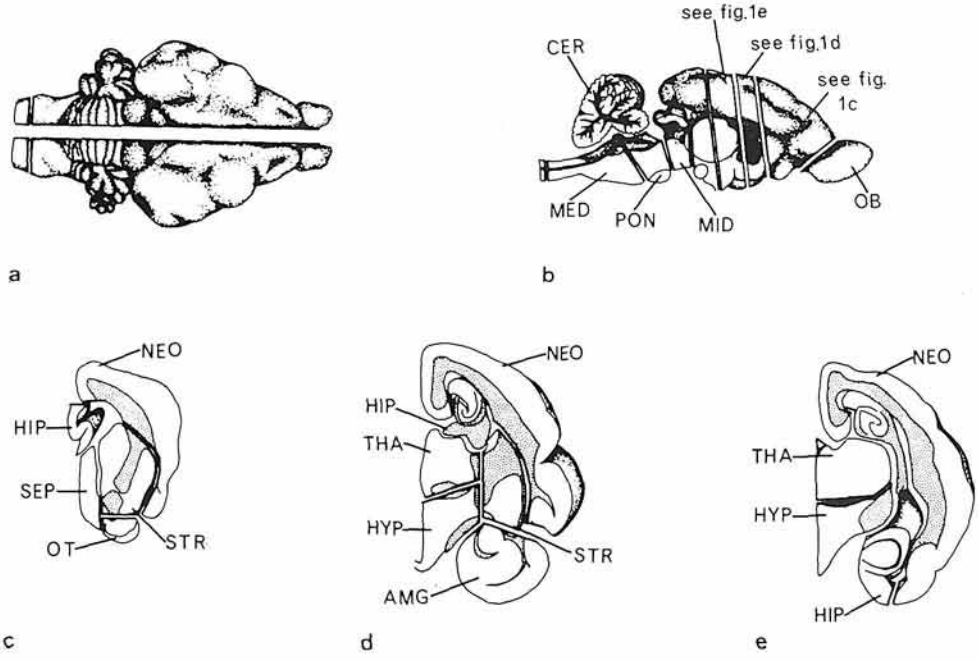


Fig. 1. The standardized set of cuts made to obtain the separately weighed segments of the brain. The brain illustrated is that of the quokka. The transverse cut illustrated in part a was made at the level of decussation of the pyramids. The sections illustrated in parts c–e are the posterior surfaces of the segments indicated in part b. AMG = Amygdala plus pyriform cortex; CER = cerebellum; HIP = hippocampus; HYP = hypothalamus; MED = medulla; MID = midbrain; NEO = neocortex; OB = olfactory bulbs; OT = olfactory tubercles; PON = pons; SEP = septum; STR = striatum; THA = thalamus.

Table I. Brain and body weights (mean ± SE)

	Body weight, kg	Whole brain weight, g	Brain/body weight ratio, g/kg
Rat	0.27 ± 0.01**	1.94 ± 0.02**	7.30 ± 0.07**
Possum	1.61 ± 0.17	8.37 ± 0.495	5.21 ± 0.31**
Rabbit	3.25 ± 0.17	9.48 ± 0.63	2.92 ± 0.195
Quokka	3.28 ± 0.14	10.15 ± 0.25	3.10 ± 0.08
Cat	3.54 ± 0.155	28.85 ± 1.555**	8.15 ± 0.44**
Sheep	45 (estimated)	126.00 ± 8.63**	2.8 (estimated)
Human	70 (estimated)	1,264.47 ± 72.06**	18.06 (estimated)

Asterisks indicate significant difference from quokka values as determined by the Student-Newman-Keuls test.

* $p < 0.05$; ** $p < 0.01$.

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Table II. Weights (g) of component parts of the brain (mean ± SE)

	Cerebellum	Medulla	Pons	Midbrain	Olfactory bulbs	Olfactory tubercles
Rat	0.28 ±0.005	0.11 ±0.005	0.11 ±0.001	0.13 ±0.003	0.04 ±0.002	0.07 ±0.002
Possum	1.33 ±0.10	0.49 ±0.05	0.32 ±0.03	0.43 ±0.03	0.24 ±0.02	0.275 ±0.015
Rabbit	1.32 ±0.15	0.515 ±0.04	0.39 ±0.025	0.79 ±0.07	0.24 ±0.03	0.27 ±0.03
Quokka	1.40 ±0.03	0.47 ±0.02	0.40 ±0.01	0.53 ±0.01	0.32 ±0.01	0.30 ±0.01
Cat	4.14 ±0.17	1.15 ±0.045	0.83 ±0.04	1.11 ±0.045	0.46 ±0.03	0.58 ±0.07
Sheep	13.98 ±0.40	3.91 ±0.19	2.80 ±0.17	4.83 ±0.205	0.98 ±0.15	2.42 ±0.18
Human	130.49 ±8.30	5.06 ±0.65	14.26 ±0.85	8.06 ±0.45	0.16 ±0.02	1.87 ±0.24

development of the amygdala. This is the first part of a series of more detailed investigations of the comparative anatomy of the amygdaloid complex. While *Stephan and Andy* [1977] investigated the role of the amygdala in the phylogenetic development of the brain, they did not include marsupials in their study.

Since body size and life-style as well as phylogeny have been argued to determine the relative development of separate parts of the brain [*Jerison*, 1973; *Gould*, 1977], the quokka brain has been compared with those of a range of eutherians. It has been compared in this respect with rabbit and sheep, which vary in size but occupy niches similar to that of the quokka, and with the rat, cat, and human from quite different niches. A comparison has also been made with a small marsupial of a life-style different from the quokka, the brush-tailed possum [*Trichosurus vulpecula*].

Materials and Methods

The brains of 10 quokkas, 10 rats, and each of 5 possums, cats, rabbits, sheep, and humans were examined. All animals, except sheep and humans, were weighed, perfused with 0.9% saline followed by 10% formol saline, and their brains removed. The cranial nerves were severed at their junction with the skull and the brain stem separated from the spinal cord by a cut at the level of the pyramidal decussation to permit removal of the brain from the cranial cavity. Par-

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Septum	Amygdala plus pyriform cortex	Hippo-campus	Thalamus	Hypo-thalamus	Striatum	Neocortex
0.02 ±0.002	0.08 ±0.003	0.13 ±0.003	0.11 ±0.003	0.03 ±0.001	0.10 ±0.004	0.58 ± 0.008
0.075 ±0.007	0.30 ±0.02	0.71 ±0.04	0.505 ±0.045	0.195 ±0.02	0.56 ±0.035	2.55 ± 0.18
0.07 ±0.01	0.24 ±0.02	0.82 ±0.05	0.61 ±0.035	0.175 ±0.03	0.51 ±0.03	3.10 ± 0.145
0.08 ±0.01	0.30 ±0.01	0.90 ±0.03	0.62 ±0.03	0.22 ±0.02	0.72 ±0.04	3.39 ± 0.12
0.16 ±0.03	0.74 ±0.03	0.965 ±0.10	1.69 ±0.145	0.46 ±0.065	1.51 ±0.135	14.39 ± 2.65
0.505 ±0.04	1.76 ±0.11	3.225 ±0.14	6.16 ±0.195	1.41 ±0.12	5.43 ±0.235	75.88 ± 3.75
0.76 ±0.10	3.42 ±0.51	11.79 ±0.72	22.07 ±2.02	2.45 ±0.22	44.545 ±2.50	996.47 ±28.41

ticular care was taken to maintain the integrity of the vulnerable olfactory bulbs, tips of the temporal lobes and flocculo-nodular lobes of the cerebellum. The sheep and human brains were removed from the skull, then fixed by immersion in 10% formalin. Brains were stored for at least 1 week in 10% formol saline before examination.

Meninges and blood vessels were peeled from the surface of the brains which were blotted dry and weighed, according to size, in a covered container on a Sartorius 2462 balance to the nearest 0.1 mg, or a Sartorius 2254 top-loading balance to the nearest 0.1 g to obtain the whole brain weight.

The neural structures were separated by cutting the brain, as illustrated in figure 1a–e, then dissecting along obvious planes of division. The segments formed were like those described by *Stephan and Pirlot* [1970] in their study of bats brains.

The weights of the separate pieces of each brain structure were added and the weights of all structures added to obtain the total brain weight which, unlike the whole brain weight, did not include choroid plexus, large blood vessels, ventricular fluid, or meningeal remnants. The whole brain weight was used in determining brain/body weight ratios and the total brain weight in determining the proportion of the brain formed by each separate structure.

Statistical comparisons were made using a one-way analysis of variance, followed by the Student-Newman-Keuls test for post hoc comparisons amongst groups of different sizes. Arc sin transformations were made of the brain/body weight ratios and of the percentage figures expressing the proportion of the total brain weight contributed by each of the separate structures, as recommended by *Sokal and Rolf* [1969].

Humans and sheep were excluded from the analysis of brain/body weight ratios, as only estimated body weights were available.

The weight of the amygdala was compared with

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Table III. Percentage distribution of total brain weight (mean ± SE)

	Cerebellum	Medulla	Pons	Midbrain	Olfactory bulbs	Olfactory tubercles
Rat	15.3 ±0.20	6.0 ±0.27	6.3 ±0.20	7.1 ±0.13**	2.3 ±0.13**	4.0 ±0.13**
Possum	16.8 ±0.65	6.1 ±0.40	4.1 +0.40	5.4 ±0.25	3.1 ±0.25*	3.5 ±0.15
Rabbit	14.0 ±0.65	5.5 ±0.20	4.2 ±0.10	8.4 ±0.10**	2.6 ±0.20**	2.9 ±0.16
Quokka	14.6 ±0.20	4.9 ±0.20	4.2 ±0.13	5.5 ±0.17	3.4 ±0.10	3.1 ±0.10
Cat	14.7 ±0.55	4.1 ±0.25	3.0 ±0.15	4.0 ±0.25**	1.6 ±0.10**	2.1 ±0.25
Sheep	11.1 ±0.20	3.1 ±0.15	2.2 ±0.20	3.8 ±0.20**	0.8 ±0.10**	1.9 ±0.10
Human	10.4 ±0.65	0.4 ±0.05	1.15 ±0.05	0.6 ±0.025	0.02 ±0.003	0.15 ±0.018

* $p < 0.05$; ** $p < 0.01$.

that of each of the other structures independently of the size of the brains to which they belonged. Partial correlation coefficients ($r_{AX\ T}$, where A represents the amygdala, X the other brain structure, and T the total brain weight) were calculated for the relationship between the weight of the amygdala and that of each other brain structure, with total brain weight considered as the fixed value in each case. The partial correlation coefficients were obtained by the method described in *Yamane* [1973].

Results

The whole brain weights and the body weights of the range of animals considered in this study are presented in table I. As would be expected, the variance in whole brain weights across all species was large as was that of the body weights.

No consistent pattern distinguishing marsupials from eutherians was apparent, either upon comparison of the whole brain weights of all species or those of the three equivalently sized species, quokka, rabbit, and cat.

A wide range in the ratios of whole brain weight to the body weight of the species was found, but the marsupials could not be distinguished on this basis from the eutherian mammals. Both quokka and possum had brain/body weight ratios which fell within the range of those of the eutherian mammals. This ratio in the

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Septum	Amygdala plus pyriform cortex	Hippo-campus	Thalamus	Hypo-thalamus	Striatum	Neocortex
1.3 ±0.10**	4.3 ±0.17**	7.3 ±0.13**	6.0 ±0.13	1.9 ±0.13	5.8 ±0.23**	32.2 ±0.27
0.9 ±0.05	3.8 ±0.25**	9.0 ±0.20	6.3 ±0.75	2.4 ±0.15	7.1 ±1.20	32.0 ±1.4
0.8 ±0.10	2.6 ±0.15*	8.8 ±0.30	6.6 ±0.25	1.8 ±0.20	5.5 ±0.20**	33.6 ±1.3
0.8 ±0.07	3.1 ±0.10	9.3 ±0.23	6.4 ±0.23	2.3 ±0.17	7.3 ±0.30	35.2 ±0.4
0.6 ±0.10	2.6 ±0.15*	3.4 ±0.30**	6.0 ±0.50	1.6 ±0.20**	5.3 ±0.35	50.7 ±2.5**
0.4 ±0.025	1.4 ±0.05**	2.6 ±0.10**	4.9 ±0.15**	1.1 ±0.15**	4.3 ±0.15**	60.1 ±1.15**
0.06 ±0.009	0.30 ±0.05	0.90 ±0.05	1.8 ±0.10	0.20 ±0.015	3.60 ±0.10	80.4 ±0.6

Table IV. Statistical relationships between species in the proportion of the total brain weight formed by structures which appeared to be more prominently represented in the marsupial brains

Striatum					
Sheep	Cat	Rabbit	Rat	Possum	Quokka
Olfactory bulbs					
Sheep	Cat	Rat	Rabbit	Possum	Quokka
Hippocampus					
Sheep	Cat	Rat	Rabbit	Possum	Quokka
Hypothalamus					
Sheep	Cat	Rabbit	Rat	Quokka	Possum

Data were taken from table III and species are arranged in ascending order of the proportion contributed by each of the named structures to the total brain weight. Bars link species which do not differ significantly ($p < 0.05$) according to the Student-Newman-Keuls test.

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Table V. Percentage distribution of non-neocortical brain weight (mean ± SE)

	Cerebellum	Medulla	Pons	Midbrain	Olfactory bulbs
Rat	22.6 ±0.33	8.9 ±0.40	9.5 ±0.37**	10.4 ±0.23**	3.4 ±0.17**
Possum	24.6 ±0.90*	9.0 ±0.60	6.2 ±0.55	8.2 ±0.40	4.5 ±0.45*
Rabbit	22.0 ±0.80	8.8 ±0.40	6.5 ±0.25	13.1 ±0.20**	4,1 ±0.35**
Quokka	22.6 ±0.23	7.6 ±0.27	6.5 ±0.23	8.5 ±0.27	5.2 ±0.13
Cat	30.0 ±1.1**	8.3 ±0.25	6.0 ±0.001	8.1 ±0.35	3.4 ±0.35**
Sheep	29.5 ±0.25**	8.2 ±0.35	5.9 ±0.40	10.0 ±0.35**	2.1 ±0.40**
Human	53.6 ±1.70	2.1 ±0.20	5.6 ±0.25	3.3 ±0.20	0.1 ±0.001

* $p < 0.05$; ** $p < 0.01$.

Table VI. Statistical relationships between species in the proportion of the non-neocortical brain weight formed by the olfactory bulbs which appeared from the data of table V to be more prominently represented in the marsupial brains

Olfactory bulbs

Sheep — Cat — Rat — Rabbit — Possum — Quokka

(Cat and Rat are linked; Rabbit and Possum are linked.)

Species arranged in ascending order of the proportion contributed by the olfactory bulbs to the non-neocortical brain weight. Bars link species which do not differ significantly ($p < 0.05$) according to the Student-Newman-Keuls test.

quokka closely resembled that of the other grazing animals, the rabbit and the sheep.

The distribution of the mass of the brains amongst their constituent structures is shown in table II. Comparison of the species can be made more easily, however, by consideration of the proportion of the total brain weight which each structure contributes (table III).

The neocortex formed the major part of

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Olfactory tubercles	Septum	Amygdala plus pyriform cortex	Hippo-campus	Thalamus	Hypo-thalamus	Striatum
6.0 ±0.20**	1.9 ±0.20**	6.3 ±0.20**	10.8 ±0.20**	8.9 ±0.17	2.9 ±0.10	8.6 ±0.40**
5.1 ±0.001	1.4 ±0.20	5.5 ±0.25	13.2 ±0.40*	9.4 ±0.55	3.6 ±0.25	10.3 ±0.85
4.5 ±0.25	1.2 ±0.20	4.1 ±0.35*	13.7 ±0.55	10.2 ±0.55	2.9 ±0.35	8.6 ±0.40
4.4 ±0.17	1.3 ±0.17	4.8 ±0.27	14.5 ±0.37	10.0 ±0.37	3.5 ±0.27	11.3 ±0.43
3.8 ±0.65*	1.2 ±0.25	5.4 ±0.25	6.8 ±0.55**	12.2 ±0.70**	3.3 ±0.25	10.9 ±0.95
5.1 ±0.35	1.1 ±0.001	3.7 ±0.25**	7.0 ±0.35**	13.0 ±0.35**	3.0 ±0.35	11.4 ±0.25
0.8 ±0.001	0.3 ±0.001	1.2 ±0.20	4.6 ±0.25	9.1 ±0.70	1.0 ±0.001	18.3 ±0.55

Table VII. Relationship between the weight of the amygdala and other brain structures

	r_{AXT}		r_{AXT}
Cerebellum	0.94**	Septum	0.90*
Medulla	0.82*	Hippocampus	0.11
Pons	0.91*	Thalamus	0.91*
Brain stem[1]	0.73	Hypothalamus	0.92**
Midbrain	−0.38	Diencephalon[2]	0.94**
Olfactory bulbs	0.81*	Striatum	0.79
Olfactory tubercles	−0.002	Neocortex	−0.65

[1] Medulla plus pons.
[2] Thalamus plus hypothalamus.
* $p < 0.05$; ** $p < 0.01$.

Table VIII. Relationship between the size of the amygdala and other brain structures

	r_{AXT}		r_{AXT}
Cerebellum	0.35	Hippocampus	0.67
Brain stem[1]	0.55	Diencephalon[2]	0.57
Midbrain	0.52	Striatum	0.40
Olfactory bulbs	0.92	Neocortex	−0.75
Septum	0.73		

Values derived from the brain volumes of the 78 non-hominoid species considered in the papers of *Stephan and Pirlot* [1970] and *Stephan* et al. [1970]. All coefficients are statistically significant ($p < 0.05$; d.f. = 76).

[1] Medulla plus pons.
[2] Thalamus plus hypothalamus.

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each of the types of brain studied, but showed the greatest variation of all structures in the proportion it contributed to the total brain weight of the different species.

Since the extent of neocortical development of the humans clearly differentiated them from all other species considered, they were not included in the analysis of the data of table III.

Marsupials did not differ from eutherians in the proportion the neocortex contributed to the total brain weight, having values similar to those of the rat and rabbit. The proportionate contribution made by most structures to the total brain weight of marsupials fell within the eutherian range. The marsupials had higher proportions of striatum, olfactory bulbs, hippocampus, and hypothalamus in their brains than most of the eutherians, but only in the proportionate size of the striatum were both marsupials significantly different from all eutherians (table IV).

The size and variability of the neocortex largely determined the variations in proportions of the smaller structures of the brain, so the sizes of the other structures were considered independently of the cortex to assess their relative development in the different species. The proportion each structure formed of the part of the brain which remains after removal of the neocortex was calculated and is shown in table V.

Human brains were again found to have a distribution of mass grossly different from all other species and were excluded from further analysis.

The marsupials did not differ significantly from the eutherians in the proportion contributed to the non-neocortical brain weight of any structure but the olfactory bulbs. Both quokka and possum had relatively large olfactory bulbs. Those of the quokka were significantly larger than those of any of the eutherians (table VI).

The amygdala of marsupials was not different from that of eutherians in the proportion of the brain it formed. It appeared to be smaller in grazing animals, including the quokka, than in those species which do not graze.

The partial correlation coefficients which describe the strength of the relationship between the weight of the amygdala and the other brain structures considered independently of total brain weight are shown in table VII. Humans were not included in the calculations.

The weight of the amygdala was found to vary directly with that of the structures of the lower brain stem, the diencephalon, the olfactory bulbs, and the septum.

Discussion

The results of this study indicate that marsupial brains do not differ distinctly in their gross morphology from eutherian brains. They fell within the range of size of the eutherian brains studied when considered relative to body size. Doubt is thus cast upon the claim of *Jerison* [1961] that marsupials have smaller brains relative to their body size than eutherians. This claim, however, was based upon data from a single American opossum [*Didelphis virginiana*]. Furthermore, the animal was selected from amongst others with higher brain/body weight ratios. It is possible that the apparently higher relative brain sizes of the eutherians of *Jerison*'s

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study were due to the particular range of animals he selected. The species studied were not identified in his paper. The brain/body weight ratios of the opossums in *Jerison*'s study (3.71 and 1.87 g/kg) are similar to those of the rabbit and quokka in my study, and values published in the works of *Stephan* et al. [1970] and *Ploog and Melnechuk* [1969] confirm the existence of eutherians with lower brain/body weight ratios than the marsupials of my study.

The direct index of relative brain size used in this study, the brain/body weight ratio, was not the same as that used by *Jerison* [1961]. *Jerison*'s index of encephalization (K) was based upon an allometric formula ($\log K = \log E - \beta \log P$, where E represents brain weight, P body weight and β the slope of the regression line = 0.67), empirically derived from the regression of brain on body weight. The numerous difficulties associated with the use of allometric indices have been described well by *Holloway* [1974]. However, the differences in the results of *Jerison*'s study and the present one were not due to the different indices employed. The indices of encephalization of the marsupials of my study (quokka = 0.053, possum = 0.069) still fell within the eutherian range (rabbit = 0.049, rat = 0.051, sheep = 0.119, cat = 0.142, human = 0.896) when the methods of calculation of *Jerison* [1961] were used.

A straightforward interpretation of the results of comparisons of brain/body weight ratios amongst species is difficult. The report of *Hodos* [1970] on the rate of acquisition of learning sets in different species provided general support for the view that greater relative brain size is associated with a greater degree of intelligence. However, allowance must be made for the influence of natural selection on body weight independently of brain weight. Ruminants, with their large guts, have brain/body weight ratios of about 2 g/kg, while bats, adapted to flight, have ratios of about 34 g/kg, twice that of humans [*Ploog and Melnechuk*, 1969; *Stephan and Pirlot*, 1970]. It is only appropriate, therefore, to compare animals with similar adaptations of body and life-style when evaluating differences in brain relative to body size. In this study the comparisons between the quokka and the grazing eutherians, especially the rabbit which resembles the quokka in both body size and morphology, are the most informative. The quokka has a slightly larger brain than the rabbit.

Marsupial brains do not differ in their basic composition from eutherian brains. In this study the proportionate contribution to the total brain weight and the non-neocortical brain weight of most structures was similar in marsupials and eutherians. It should be noted particularly that the proportion of the brain formed by the neocortex of the marsupials lies within the eutherian range. Furthermore, the neocortex formed a greater part of the brains of the marsupials of this study than of all the insectivores in the study of *Stephan* et al. [1970]. Marsupials do not have neocortically underdeveloped 'primitive' brains when compared with eutherians.

The results of this study did appear to support to some extent another of the assumptions about marsupial brains, in that the olfactory bulbs were shown to be larger in these brains than in the eutherians. The's measurements of *Stephan* et al.

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[1970], however, show that some eutherians do have a greater part of their non-neocortical brain formed by the olfactory bulbs than the marsupials of my study. Furthermore, my study showed no enlargement of other rhinencephalic structures in marsupials compared with eutherians. The olfactory tubercle, septum, hippocampus, and amygdala have all been considered to be part of the rhinencephalon, and yet formed no greater part of the brain of the marsupials than of the eutherians. It cannot be said that marsupial brains are characterized by an outstanding rhinencephalic development.

It has been argued that ontogenetic or functional relationships between structures are reflected in consistent relationships between the sizes of such structures in brains of different size and composition [see, e.g., *Stephan and Andy,* 1977]. The size of the amygdala was found to be related to the size of the structures which project to it (the lower brain stem structures), to the size of the structures to which it projects (the diencephalon), and to the size of the structures with which it has reciprocal connections (the septum and olfactory bulbs).

Both *Stephan and Andy* [1977] and *Breathnach and Goldby* [1954] have rejected the idea that amygdaloid size is related to that of the olfactory bulbs and have argued for its developmental and functional independence from the rhinencephalon. The study of *Breathnach and Goldby* [1954] was based upon single samples of very few species and, therefore, carries little statistical weight. *Stephan and Andy* [1977] used a wide range of species, but their conclusions were based upon simple correlation coefficients obtained by direct comparison of absolute volumes of the various parts of the brain. Such indices are not appropriate, however, as the relationships between the parts of the brain are obscured by the stronger relationship between each of the parts and the whole brains from which it was taken. Partial correlation coefficients, on the other hand, show the strength of the correlation between the sizes of the various parts independently of the correlation between the parts and the whole brains. Partial correlation coefficients could not be calculated from the data presented by *Stephan and Andy* but could be obtained from data presented in *Stephan*'s two earlier studies which included the species studied by *Stephan and Andy* [1977]. The partial correlations for the size of the amygdala compared with other brain structures independently of whole brain size for the non-hominoid species studied by *Stephan* are shown in table VIII. They confirm the close relationships between the size of the amygdala and the olfactory bulbs demonstrated in this study.

The data of *Stephan* also confirm the relationship between the size of the amygdala and septum, and the inverse relationship between the sizes of the amygdala and neocortex only suggested by the present study.

Within the confines of my study, marsupial brains have been shown not to differ in gross morphology from those of eutherians. However, the lack of difference cannot be said to demonstrate similarity of function between marsupial and eutherian brains. While gross morphology provides a basic indication of brain function, factors such as the cellular composition, synaptic complexity, chemical speci-

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ficity, and metabolic sensitivity of brains must obviously have important influences upon brain function. The way in which marsupials compare with eutherians in these matters is yet to be determined.

References

Breathnach, A.S.; Goldby, F.: The amygdaloid nuclei, hippocampus and other parts of the rhinencephalon in the porpoise (*Phocaena phocaena)* J. Anat. *88:*267–291 (1954).

Gould, S.J.: Ontogeny and phylogeny (Belknap Press, Cambridge 1977).

Hodos, W.: Evolutionary interpretation of neural and behavioral studies of living vertebrates; in Schmitt, The neurosciences second study program, pp. 26–39 (Rockefeller University Press, New York 1970).

Holloway, R.L.: On the meaning of brain size. Science *184:*677–679 (1974).

Jerison, H.J.: Quantitative analysis of evolution of the brain in mammals. Science *133:* 1012–1014 (1961).

Jerison, H.J.: Evolution of the brain and intelligence (Academic Press, New York 1973).

Ploog, D.; Melnechuk, T.: Primate communication. A report based on an NRP work session. Neurosci. Res. Progm Bull. *7:*468–473 (1969).

Smith, G.E.: The cerebral commissures of the mammalia, with special reference to Monotremata and Marsupialia. Proc. Linn. Soc. NSW *9:* 635–657 (1894).

Sokal, R.R.; Rohlf, F.J.: Biometry. The principles and practice of statistics in biological research, pp. 380–387 (Freeman, San Francisco, 1969).

Stephan, H.; Andy, O.J.: Quantitative comparison of the amygdala in insectivores and primates. Acta anat. *98:* 130–153 (1977).

Stephan, H.; Bauchot, R.; Andy, O.J.: Data on size of the brain and of various brain parts in insectivores and primates; in Noback, Montagna, The primate brain. Adv. Primatol. *1:*289–297 (1970).

Stephan, H.; Pirlot, P.: Volumetric comparisons of brain structures in bats. Z. zool. Syst. Evolforsch. *8:*200–236 (1970).

Tyndale-Biscoe, H.: Life of marsupials (Arnold, London 1973).

Yamane, T.: Statistics: an introductory analysis (Harper & Row, New York 1973).

Young, J.Z.: The life of vertebrates (Oxford University Press, New York 1962).

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